# C1 - FIR-QP 俯仰扰动恢复详细流程

**离线辨识 / Offline identification**

FDDP nominal + C0-aligned pitch reference
$q_{ref}(t),\quad \theta_{ref}(t)$
固定相位 C1 reference

↓

经验脉冲响应辨识
$\alpha$ *pulse → root-pitch deviation*
$h = [h_0,\ldots,h_{M-1}]$
得到 FIR response coefficients

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拟合并稳定化 AR disturbance model
$d_k = \Sigma a_j d_{k-j}$
检查 / 收缩 spectral radius

↓

**在线 Receding-Horizon Control**

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测量俯仰并形成参考对齐误差
$e_k = \theta_{meas,k} - \theta_{ref}(t)$

↓

减去历史指令的已知延迟响应
$d_k = e_k - \Sigma h_i \alpha_{k-i}$

↓

预测自由响应并构造 FIR dynamic matrix
$E_{free} = E_{AR} + E_{past}$
$E = E_{free} + SA$

↓

求解 tracking / magnitude / slew QP
$A^* = arg\ min\ q\|E_{free}+SA\|^2 + r\|A\|^2 + s\|DA\text{-}d_{prev}\|^2$
$|A_i| \le \alpha_{max},\quad |\Delta\alpha| \le \Delta\alpha_{max}$

↓

仅执行优化序列的第一个元素
$\alpha_k = A_0^*$

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映射为关节修正与位置命令
$\Delta q_k = G\alpha_k$
$q_{cmd} = q_{ref}(t) + \Delta q_k$

↓

计算并检查 acceptance metrics
IAE / peak & final pitch error / min contacts
$max\ |\Delta q|$
/ median & max QP solve time

证据边界：C1 相对 C0 改善扰动后的恢复性能；它不声称消除扰动施加瞬间的初始峰值。